

Network Health & Troubleshooting Assessment

1. Executive Summary

This diagnostic report assesses the operational integrity of the NetSage virtual enterprise network environment. The assessment was performed under the simulation scenario 'OSPF MD5 Authentication Mismatch on R1-R2 Link'.

Currently, 4 out of 7 devices are reporting Healthy status. There are 1 open active network incidents that require immediate operational review. 1 historical incidents have been cataloged and resolved.

2. Device Status Matrix

Device Name	Type	IP Address	Health Status	CPU / Mem Usage	Last Checked
R1	Router	192.168.12.1	Critical	8.0% / 42.0%	07:53:07
R2	Router	192.168.12.2	Critical	7.0% / 39.0%	07:53:07
R3	Router	192.168.23.2	Critical	6.0% / 38.0%	07:53:07
Switch1	Switch	192.168.1.2	Healthy	0.0% / 0.0%	07:53:07
Switch2	Switch	192.168.4.2	Healthy	0.0% / 0.0%	07:53:07
Server01	Server	192.168.4.10	Healthy	0.0% / 0.0%	07:53:07
Client01	Client	192.168.1.50	Healthy	0.0% / 0.0%	07:53:08

3. Active Incident Diagnostics & Recommendations

Incident INC-002 | R1 | Severity: High (Confidence: 100%)

Issue Type: OSPF Adjacency Failure due to Authentication Mismatch

Root Cause: OSPF Authentication Mismatch

AI Technical Advisor Analysis:

Root Cause: OSPF Adjacency state is DOWN between router R1 and R2 due to a Cryptographic Message-Digest Authentication Mismatch. R1 is configured with MD5 key 'CISCO123', while R2 is configured with the incorrect password 'WRONG_KEY'.

Impact Analysis: OSPF neighbor state cannot reach FULL. Routers drop OSPF Hello packets due to invalid signature, stopping all routing propagation between Head Office and Branch networks.

Recommended Remediation: Align the OSPF message-digest authentication key on both connecting interfaces to use identical passwords.

- Access R2 configuration mode.
- Select interface: 'interface GigabitEthernet0/0'.
- Modify the md5 key: 'ip ospf message-digest-key 1 md5 CISCO123'.
- Ensure authentication is enabled: 'ip ospf authentication message-digest'.

Validation Commands to Run:

> show running-config interface GigabitEthernet0/0

> show ip ospf neighbor